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MongoDB Health Check — Key Steps for Optimal Performance

When performing a **MongoDB health check**, I follow these key steps to ensure optimal stability, data consistency, and performance:

1. **Verify MongoDB Services**
   - Confirm mongod, mongos, and config server processes are running.
   - Check service status and startup configuration on all nodes.
2. **Review MongoDB Logs**
   - Inspect the mongod.log for warnings, slow queries, replication lag, or connection errors.
   - Review system logs for OS-level issues impacting MongoDB.
3. **Check Cluster & Replica Set Status**
   - Run rs.status() to verify primary/secondary state, replication health, and member sync status.
   - Ensure there is no replication lag or election instability.
4. **Assess Sharded Cluster Health (if applicable)**
   - Run sh.status() to confirm all shards, config servers, and mongos routers are connected.
   - Validate chunk distribution and balancer state.
5. **Monitor Resource Utilization**
   - Review CPU, memory, disk I/O, and network metrics.
   - Identify high resource consumption or swapping.
6. **Evaluate Storage & Disk Usage**
   - Monitor data directory size and free space.
   - Verify journaling is enabled and disk latency is within acceptable limits.
7. **Check Database and Collection Statistics**
   - Use db.stats() and db.collection.stats() to review storage size, index size, and document count.
   - Identify collections with abnormal growth.
8. **Index and Query Performance**
   - Use explain() to analyze query execution plans.
   - Detect missing or unused indexes and long-running operations via db.currentOp().
9. **Replication & Oplog Health**
   - Validate oplog size and ensure sufficient retention window.
   - Check replication delay and heartbeat latency between nodes.
10. **Backup & Restore Validation**
    - Confirm that backup jobs (mongodump, snapshot, or cloud backups) are recent and restorable.
    - Test restore to staging environment periodically.
11. **Security & Access Review**
    - Verify authentication (SCRAM-SHA), user roles, and privileges.
    - Review IP bindings, TLS/SSL setup, and audit logs for failed access attempts.
12. **Configuration Review**
    - Validate key parameters like wiredTigerCacheSizeGB, storageEngine, replication.oplogSizeMB, etc.
    - Check for any deviations from baseline configurations.
13. **Monitor Alerts & Metrics**
    - Review dashboards in MongoDB Atlas, CloudWatch, or custom Prometheus/Grafana setups.
    - Ensure alert thresholds (replication lag, disk space, memory) are tuned appropriately.

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14. **Run Validation & Consistency Checks**
    - Use db.collection.validate() for data integrity.
    - Verify schema consistency if using schema validation or MongoDB Compass.
15. **Document Findings & Actions**
    - Record all checks, detected issues, and corrective actions.
    - Update runbooks and health check logs for future audits.